

STA304: Learning Activity 1 Sample Solutions

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Part 2: Learning Activity Solutions

Question 1: Sampling Elements

Target Population: All students enrolled in STA304 during the Fall 2023 semester.

Observation Unit: An individual student enrolled in the course.

Sampling Frame: Example: The official Registrar's list of active UofT student emails for the course.

Sampling Unit: A student email.

Question 2: Selection Bias Analysis

Procedure: Surveys are distributed and collected electronically via the university's course evaluation system.

Possible Sources of Selection Bias:

- **Nonresponse Bias:** The response rate was approximately 22% (51 out of 229). If nonrespondents have different views than respondents, the results are biased.
- **Self-Selection Bias:** Participation is voluntary. Often, students with strong opinions or high personal interest in the course are more likely to respond.

Justification: Selection bias is highly likely. Because such a large percentage of the population (78%) did not respond, we cannot assume the sample is representative of the whole class; it likely skews toward students with more extreme experiences.

Question 3: Questionnaire Design - Question 3

Concept of Interest: Student perception of the learning environment/classroom climate.

Issue: Vague Language. Terms like "atmosphere" and "conducive" are subjective and may be interpreted differently by each respondent.

Revision: "Did the instructor encourage students to ask questions during class?"

Question 4: Questionnaire Design - Question 4

Concept of Interest: The educational value of course assessments.

Issue: Double-Barreled Structure. The question asks about projects, assignments, tests, and exams in a single item. A student might find assignments helpful but exams unhelpful, making a single response ambiguous.

Revision: “The course assignments improved my understanding of the material” (Answered on the 1-5 scale).

Question 5: Leading Questions

Leading Revision: “How helpful was the high-quality assistance provided by the teaching staff in supporting your learning?”

Explanation: This is leading because it uses loaded adjectives (“high-quality,” “helpful”) that presuppose a positive experience and nudge the respondent toward a favorable answer.

Question 6: Reducing Measurement Error

The survey uses several design strategies to reduce measurement error:

- **Anonymity:** Reduces social desirability bias by ensuring students feel safe providing honest (even negative) feedback.
- **Closed Questions:** Standardized Likert scales (1-5) minimize the variation in how answers are interpreted compared to open-ended text.
- **Timing of survey:** The survey is administered before the final grades are released so students are not influenced by their final grades in the course when completing the survey.
- **Publication of survey results:** The survey results are not released to the instructor until after the final grades are released on acorn. This further helps in ensuring students feel safe providing honest (even negative) feedback, minimizing measurement bias.

Question 7: Reflection on Accuracy

Student evaluations may not always provide an accurate measurement of satisfaction due to selection biases. While they are useful for internal benchmarks, high nonresponse rates and self-selection mean they often lack the representativeness of a true probability sample. Additionally, if questions are vague or double-barreled (as seen in Questions 3 and 4), measurement error increases, further distancing the statistics from the true population parameters.